The triangular case of the Moving Sofa Problem

What is the maximum-area triangle that can be moved through a 90° *corner of unit width?*

Subject: Mathematics

Word Count: 3943

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1 Introduction

The inspiration for this investigation appeared during a math lesson about optimization. I faced a problem which asked for the longest segment that could complete a $90°$ turn inside a hallway of width 1. Although you could move and rotate the segment in any direction and the question seemed very hard to tackle, there turned out to be a precise answer. This problem boosted my curiosity and I wanted to investigate something similar. Therefore, when I discovered the Moving Sofa Problem, reading on the news that Jineon Baek [1] claimed to prove a solution, I decided to explore this topic.

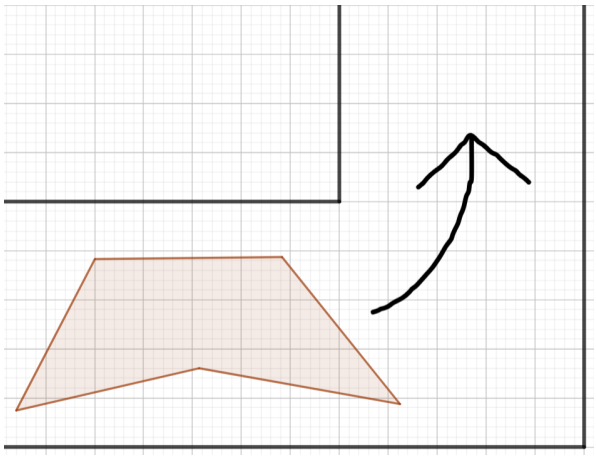


Figure 1: Visualization of The Moving Sofa Problem

The Moving Sofa Problem asks for "The largest area region which can be moved through a hallway of width 1" [2], as presented on the figure 1. There are many papers diving in advance math on the general case, for that reason we will only consider triangle shaped sofas. Although the triangle is the simplest polygon, there is an infinite number of possible paths for an infinite number of triangles. The triangle is chosen for all the known properties about it, which we could apply on its continuous movement. Thus, the research question is: "What is the maximum-area triangle, that can be moved through a $90°$ corner of unit width?". For our original investigation of the triangle case, we will search the upper and the lower bounds for the area of biggest triangle. We will proceed with an optimized analytical approach which finds the biggest area triangle. We will end looking at a simulation to find the shape relying on another approach and compare the results.

Variables

On the figure 2 we can visualize the variables defined in the table 1. Apart from the general case of the Moving Sofa Problem, the variables also include the two particular cases of the problem that we will study.

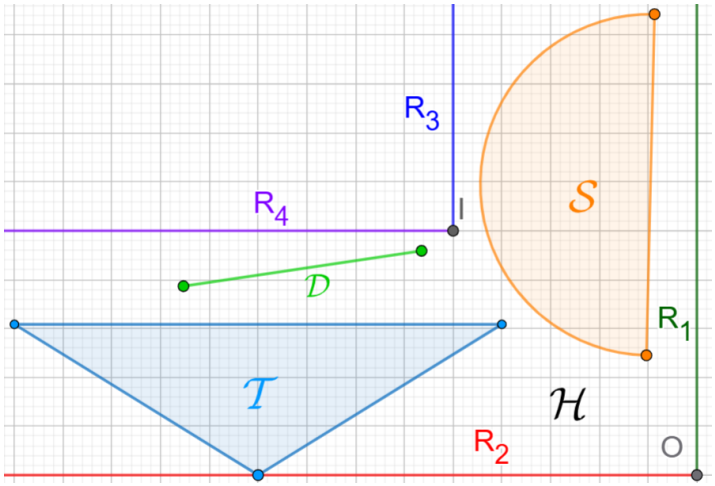


Figure 2: General definitions

Variable	**Definition**
$\mathcal{H}$	The hallway inside of which we are going to move our shapes.
O	The point on the outer corner of $\mathcal{H}$.
I	The point on the inner corner of $\mathcal{H}$.
R_1	The vertical ray with the endpoint at O pointing upwards.
R_2	The horizontal ray with the endpoint at O pointing to the left.
R_3	The vertical ray with the endpoint at I pointing upwards.
R_4	The horizontal ray with the endpoint at I pointing to the left.
$\mathcal{D}$	A one-dimensional sofa consisting of a single segment.
$\mathcal{S}$	A general two-dimensional sofa that encloses an area.
$\mathcal{T}$	Any sofa $\mathcal{S}$ that is triangle shaped.

Table 1: Variables and definitions

Minimum angle of rotation

Before considering specific sofas, we need to analyze the general characteristics of their rotation. Lets define α as the angle which rotates $\mathcal{S}$ throughout the turn. For example for a square of side 1, $\alpha = 0$ as shown on figure 3. And for a semicircle of radius 1, $\alpha = 90°$ as shown on the figure 4. Initially, $\mathcal{S}$ fits in a horizontal strip of $\mathcal{H}$. And after rotating any sofa $90°$, it will fit inside the vertical strip of $\mathcal{H}$. For this reason, we will not analyze the cases where $\alpha > 90°$. Thus, the domain of α is $[0°, 90°]$.

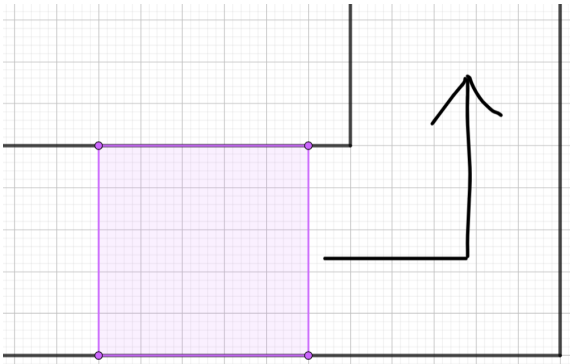


Figure 3: A sofa $\mathcal{S}$ which rotates $0°$ to complete the turn

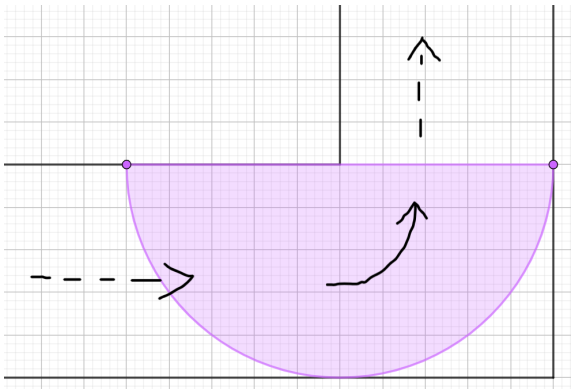


Figure 4: A sofa $\mathcal{S}$ which rotates $90°$ to complete the turn

The big difficulty of the Moving Sofa Problem is that there is no way to demonstrate that the biggest sofa must rotate the whole $90°$ to move around $\mathcal{H}$. Nevertheless, Hammersley came up with a clever way to find the minimum angle of rotation for the biggest sofa $\mathcal{S}$. This is explained in the Appendix A. Although Hammersley never published a paper talking about this, Gerver mentions this idea in his work [3].

Assumption 1

Throughout this exploration, we will assume that the biggest sofas $\mathcal{S}$, $\mathcal{D}$ and $\mathcal{T}$ will rotate exactly $90°$ to complete the turn. It is not necessary for some shapes, as we have just seen, but the biggest shapes for the analyzed cases satisfy this condition.

Note. The following sections present the author's personal investigation on the one-dimensional and triangular cases of the Moving Sofa Problem.

2 One-dimensional Moving Sofa Problem

The main objective of this investigation is to find the triangle-shaped sofa $\mathcal{T}$ of biggest area which completes the turn around $\mathcal{H}$. But before studying this special case, it will be useful to find the longest sofa $\mathcal{D}$ that completes the turn. The sofa $\mathcal{D}$ is a segment and we will maximize its length. For this case only, we will generalize the widths of the vertical and horizontal strips of $\mathcal{H}$ to a and b instead of fixing it at 1. This generalization will reveal the actual structure of the solution beyond simply finding the answer to the case $a = b = 1$. The statement is shown on the figure 5.

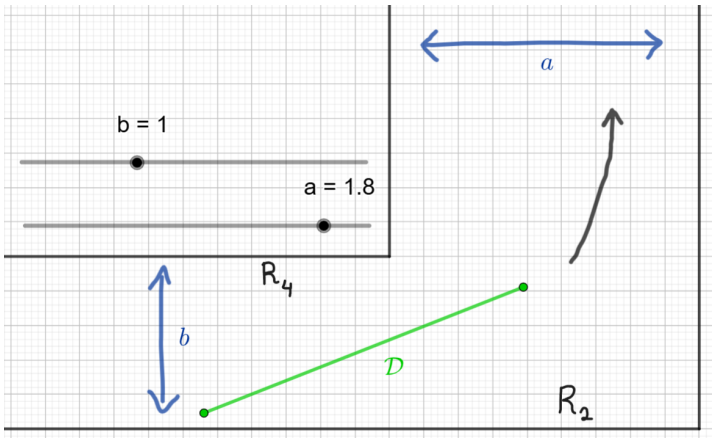


Figure 5: The initial approach for the one-dimensional sofa $\mathcal{D}$

2.1 Geometric setup

Initially, $\mathcal{D}$ is a horizontal segment parallel to R_2 and R_4. We can define an angle β showing how much has rotated $\mathcal{D}$ with respect to its initial position. In other words, the angle between $\mathcal{D}$ and R_4. As we have assumed in 1, $\mathcal{D}$ will rotate $90°$. Therefore, β will pass through all the angles in between: $\beta \in (0°, 90°)$.

Observation 1

At any moment during the rotation of $\mathcal{D}$ we can translate $\mathcal{D}$ upwards so it touches I and is entirely inside of $\mathcal{H}$. The operation is shown on the figure 6.

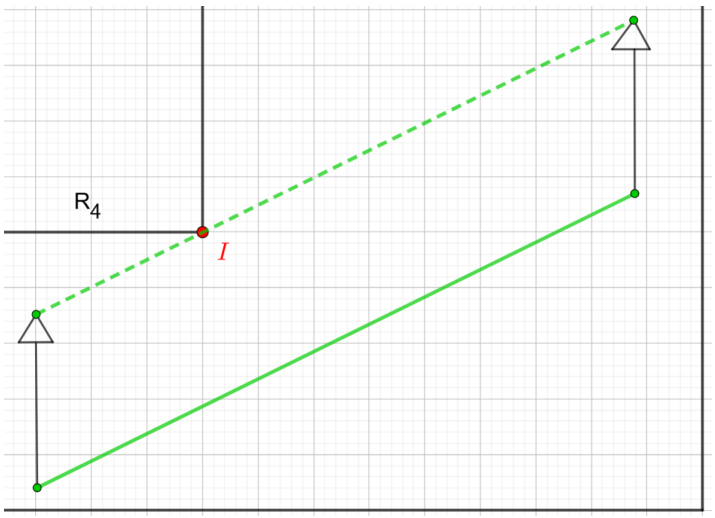


Figure 6: The translation of $\mathcal{D}$

Applying the observation 1 at each moment during the rotation of the sofa, we can only consider the positions of the segment $\mathcal{D}$ when $I \in \mathcal{D}$. Therefore, $\mathcal{D}$ is a rotating segment which continuously stays in contact with I. During this rotation the only changing variable is β.

For each possible angle β of rotation, we can extend the segment $\mathcal{D}$ to a longer segment $\overline{AB}$ of length L, where A lies on R_2 and B lies on R_1. Lets define the function $L(\beta)$ which calculates L in function of the angle β. From the figure 7 we can see that:

$$L(\beta) = \overline{AB} = \overline{AI} + \overline{IB} = \frac{b}{\sin(\beta)} + \frac{a}{\cos(\beta)}$$

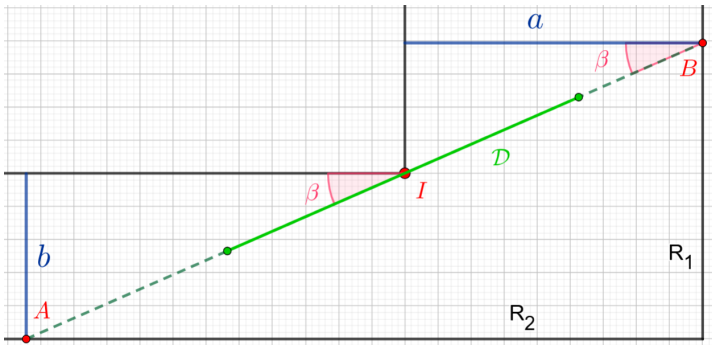


Figure 7: Trigonometric approach to find $\overline{AB}$

2.2 Optimization

$\mathcal{D}$ will move through all angles β. L is the maximum length of $\mathcal{D}$ for each angle β. Therefore, when $L(\beta)$ is minimum, $\mathcal{D}$ faces the strongest restriction. For that reason, the longest sofa $\mathcal{D}$ possible is

$$\mathcal{D}_{max} = \min(L(\beta))$$

In order to find the minimum of this function we should solve $L'(\beta) = 0$ and verify that it is a minimum.

Taking the derivative with respect to β, we get:

$$L'(\beta) = b \cdot \left(-\frac{\cos(\beta)}{\sin^2(\beta)}\right) + a \cdot \left(\frac{\sin(\beta)}{\cos^2(\beta)}\right)$$

To find the critical points, we set $L'(\beta) = 0$:

$$a \cdot \frac{\sin(\beta)}{\cos^2(\beta)} = b \cdot \frac{\cos(\beta)}{\sin^2(\beta)}$$

Multiplying both sides by $\sin^2(\beta)\cos^2(\beta)$, we obtain:

$$b\cos^3(\beta) = a\sin^3(\beta)$$

Solving for $\beta \in (0°, 90°)$, we find:

$$\beta_{min} = \arctan\left(\sqrt[3]{\frac{b}{a}}\right)$$

Minimum check

To verify whether $L(\beta)$ is concave up or concave down lets rewrite it as:

$$L(\beta) = b\csc(\beta) + a\sec(\beta)$$

Notice that $\sin x$ and $\cos x$ are concave down on the range $(0, \pi/2)$. For that reason, their

reciprocals $\csc x$ and $\sec x$ respectively, must be concave up [4]. The result of positive linear combinations of functions that are concave up, is another function that is concave up [4]. Hence, β_{min} is the minimum.

2.3 Solution

We can substitute β_{min} back to $L(\beta)$ to find the longest possible segment $\mathcal{D}_{max}$:

$$L\left(\arctan\left(\sqrt[3]{\frac{b}{a}}\right)\right) = \frac{b}{\sin\left(\arctan\left(\sqrt[3]{\frac{b}{a}}\right)\right)} + \frac{a}{\cos\left(\arctan\left(\sqrt[3]{\frac{b}{a}}\right)\right)}$$

Lets define

$$\theta = \arctan\left(\sqrt[3]{\frac{b}{a}}\right)$$

When dealing with a trigonometric function inside another trigonometric function drawing a triangle like on the figure 8 can be useful. Applying the Pythagorean theorem we find its hypotenuse:

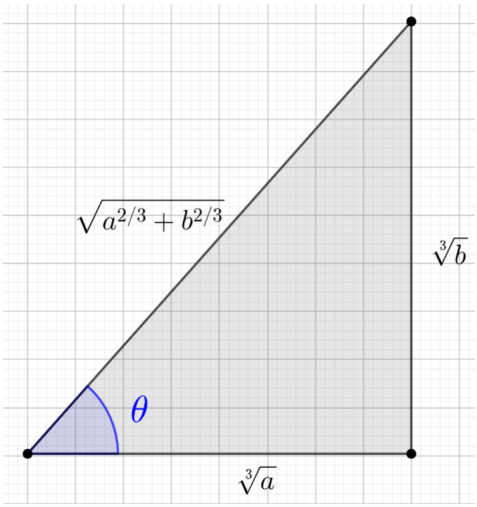


Figure 8: Trigonometric simplification

This way,

$$\begin{cases} \cos(\theta) = \dfrac{\sqrt[3]{a}}{\sqrt{a^{2/3}+b^{2/3}}}, \\ \sin(\theta) = \dfrac{\sqrt[3]{b}}{\sqrt{a^{2/3}+b^{2/3}}} \end{cases}$$

$$\mathcal{D}_{max} = \frac{b\sqrt{a^{2/3}+b^{2/3}}}{\sqrt[3]{b}} + \frac{a\sqrt{a^{2/3}+b^{2/3}}}{\sqrt[3]{a}}$$

$$= b^{2/3}\sqrt{a^{2/3}+b^{2/3}} + a^{2/3}\sqrt{a^{2/3}+b^{2/3}} = (b^{2/3}+a^{2/3})\sqrt{a^{2/3}+b^{2/3}}$$

$$= \boxed{(b^{2/3}+a^{2/3})^{3/2}}$$

When $a = b = 1$,

$$\mathcal{D}_{max} = L(\beta) = (1+1)^{3/2} = 2\sqrt{2} \tag{1}$$

The solution to this problem is symmetric on a and b. This makes sense, as switching a and b, is equivalent to moving the sofa backwards after it completes the turn. And moving the sofa forward or backwards does not affect the answer.

3 The triangular Moving Sofa Problem

In this section we will analyze the triangle-shaped sofas $\mathcal{T}$. As it is unclear which approach we should use to find the biggest-area sofa $\mathcal{T}$, we will begin finding the upper and the lower bounds.

3.1 Upper and lower bounds

The first upper bound is inspired on the upper bound in "Moving a food trolley around a corner" [5].

Claim 1. *The area of any sofa $\mathcal{T}$ is smaller or equal to the area of the isosceles right triangle $T_1 = \triangle ONM$ shown in Figure 9.*

Proof

Before the turn, $\mathcal{T}$ is inside the horizontal strip of $\mathcal{H}$ outside T_1 and after the turn $\mathcal{T}$ is inside the vertical strip of $\mathcal{H}$ and outside T_1. $\mathcal{T}$ is a triangle, so it is a convex shape. The segment connecting two points inside of a convex shape lies entirely inside of the convex shape [6].

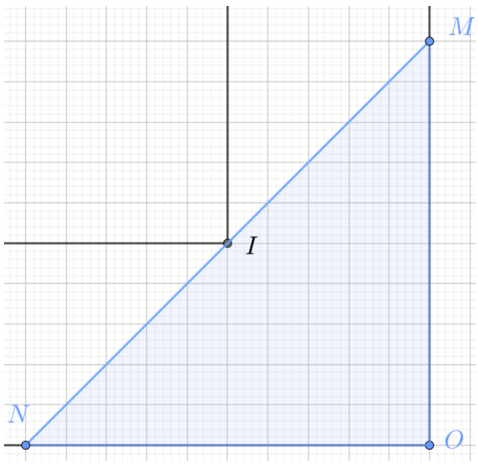


Figure 9: Triangle $T_1 = \triangle ONM$

Therefore, if at any moment during the turn $\mathcal{T}$ has at least one point P_1 in the horizontal strip of $\mathcal{H}$ on the left of $\overline{NI}$ and at least one point P_2 on the vertical strip of $\mathcal{H}$ on top of $\overline{IM}$, then the segment $\overline{P_1P_2}$ connects to points which are located inside $\mathcal{T}$. But, $\overline{P_1P_2}$ intersects the line OI outside of $\mathcal{H}$, therefore outside of $\mathcal{T}$, as we can observe on the figure 10.

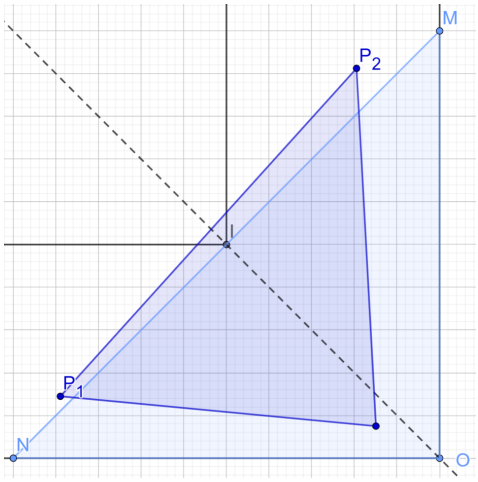


Figure 10: $\overline{P_1P_2}$ and OI intersect outside $\mathcal{H}$

This is a contradiction. Therefore, before $\mathcal{T}$ enters the vertical strip of $\mathcal{H}$ above $\overline{IM}$, $\mathcal{T}$ must be located entirely inside of T_1. Therefore, Area$(\mathcal{T}) \leq$ Area(T_1).

Q.E.D.

Area$(T_1) = \dfrac{2 \cdot 2}{2} = 2$, so Area$(\mathcal{T}) \leq 2$.

This upper bound is too general, as it works for any convex sofa. Lets come up with something particular for the triangular sofa $\mathcal{T}$.

Claim 2. *Suppose we have the sofa triangle $\mathcal{T} = \triangle ABC$ inside the horizontal strip of $\mathcal{H}$, between R_2 and R_4. Assume that B is the vertex that has one vertex A located on the left and one vertex C on the right. If two points are aligned on the same vertical line, define one of them B and the other one A. The height h_B from the vertex B is always smaller or equal to* 1.

Proof

Draw the circle c_1 with radius 1 and center B. If A is on the same vertical line as B, $|AB| \leq 1$, so A is inside c_1. If not, A and C are on one side of B each. In both cases, the line AC intersects c_1 in at least 1 point. For that reason the perpendicular from B to AC, $h_B \leq 1$, as shown on the figure 11.

Q.E.D.

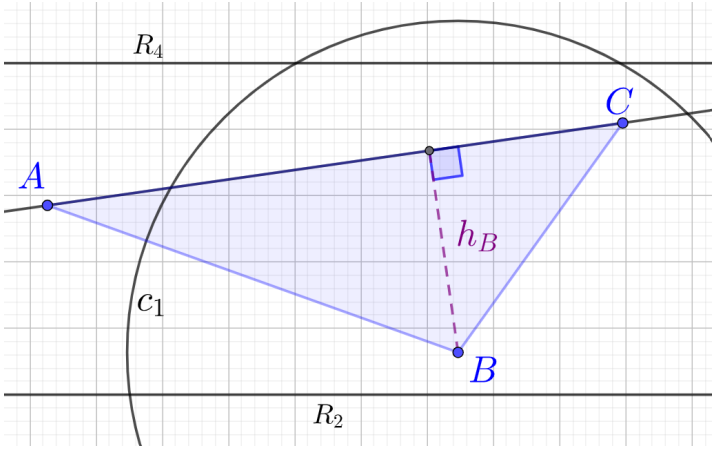


Figure 11: Height from B smaller or equal to 1

Corollary 1

As the longest segment $\mathcal{D}$ we can move around the corner is $2\sqrt{2}$ by the formula 1, we know that the 3 sides of $\mathcal{T}$ are smaller or equal to $2\sqrt{2}$. Also, one of the heights is smaller or equal to 1. So, the maximum area of $\mathcal{T}$ is $\dfrac{2\sqrt{2} \cdot 1}{2} = \sqrt{2} \approx 1.414$.

Lower bound

For the lower bound, lets find an example of a sofa $\mathcal{T}$ that can navigate the turn. A logical way to turn $\mathcal{T}$ is to rotate it around I. This way, we can build the sofa $\mathcal{T}$ from the figure 12 which can move around $\mathcal{H}$ rotating $90°$.

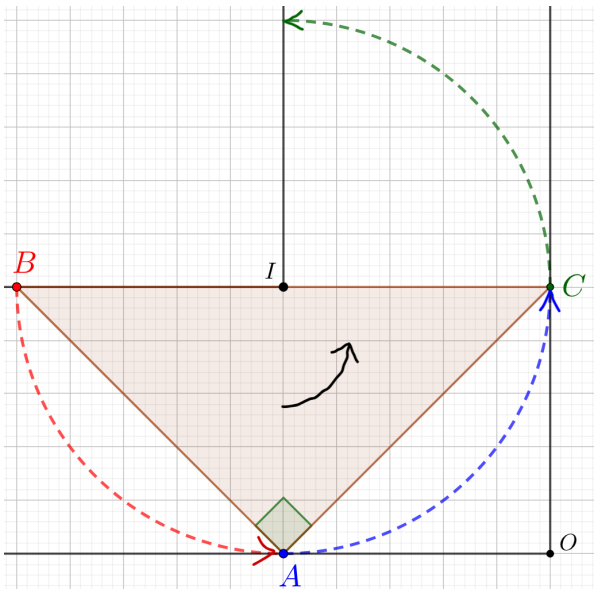


Figure 12: An example of $\mathcal{T}$ rotating points around I

The length of the hypotenuse $\overline{BC}$ of this triangle is 2 and the height h_A is 1. Thus, $\text{Area}(\triangle ABC) = \dfrac{2 \cdot 1}{2} = 1$. We have narrowed the range of values for the maximum area of a sofa $\mathcal{T}$ that can move around $\mathcal{H}$ to $[1, \sqrt{2}]$.

Remark 1

Let $\mathcal{T}_1$ be a sofa that can move around $\mathcal{H}$. Define $\mathcal{T}_2$ as the symmetric shape to $\mathcal{T}_1$ with respect to the line IO. The corner $\mathcal{H}$ is also symmetric with respect to IO. Hence, during all the rotation, the image is completely symmetric about IO. As $\mathcal{T}_1$ can complete the turn around $\mathcal{H}$, $\mathcal{T}_2$ can also. The figure 13 shows $\mathcal{T}_1$ and $\mathcal{T}_2$ during their turn around $\mathcal{H}$. As $\mathcal{T}_1$ and $\mathcal{T}_2$ are equivalent, we can only consider one of them.

Now, we should distinguish 2 different cases in order to restrict the initial position of the sofas $\mathcal{T}$:

1. $\mathcal{T}$ has the side $\overline{AC}$ lying on R_2 before starting the turn.

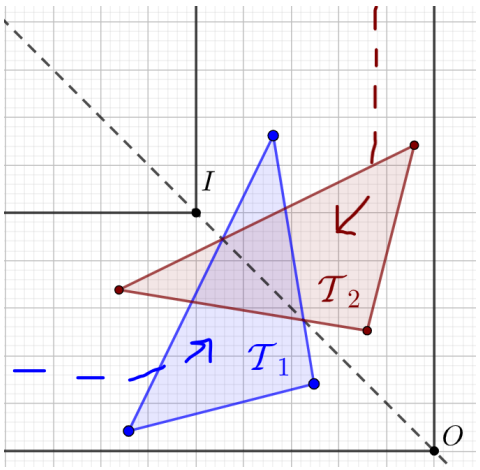


Figure 13: Symmetric movements of $\mathcal{T}_1$ and $\mathcal{T}_2$ with respect to IO

2. $\mathcal{T}$ has the side $\overline{BC}$ lying on R_4 before starting the turn.

Corollary 2

By claim 2, at least one height of $\mathcal{T}$ is less or equal to the width of $\mathcal{H}$. Hence, all the triangles inside the horizontal strip of width 1 can be rotated to satisfy one of the 2 cases presented on the figure 14.

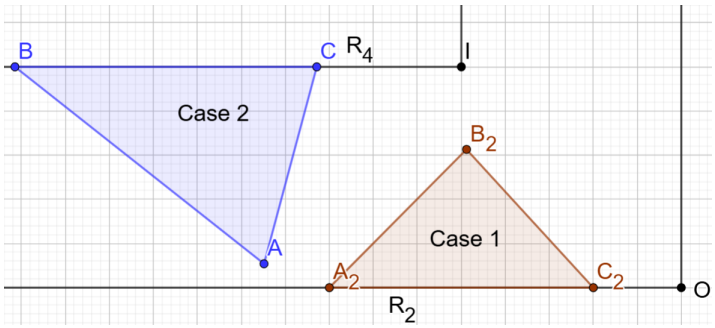


Figure 14: The two possible cases before starting moving the sofa

3.2 Case 1

Claim 3. *For each acute triangle satisfying case* 1 *which can be moved around* $\mathcal{H}$*, there exists a right angle triangle of same area that also can move around* $\mathcal{H}$*.*

Proof

By assumption 1, the biggest area triangle sofa will rotate $90°$. Hence, for the first case, lets look at the acute triangle $\mathcal{T}$ after it completes a $45°$ turn. By remark 1, for this rotation $\mathcal{H}$ is symmetric about IO. If the $\triangle ABC$ from the figure 15 can move around $\mathcal{H}$, then the right trapezoid $ABB'C$ can also make the turn.

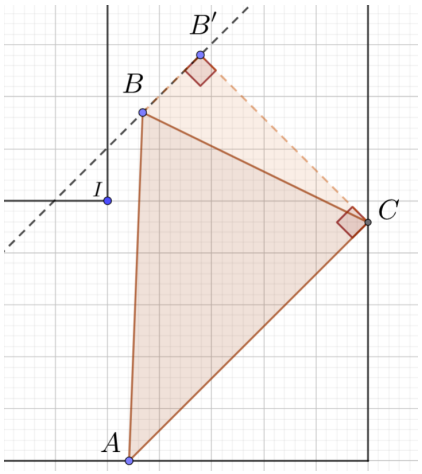


Figure 15: Acute triangle from case 2 contradiction

The right angle triangle $\triangle AB'C$ is inside the trapezoid $ABB'C$, therefore it will also complete the turn. As Area$(\triangle AB'C)$ = Area$(\triangle ABC)$, for each acute triangle $\triangle ABC$ we can find a right angle triangle $\triangle AB'C$ of same area that also can move around $\mathcal{H}$.

Q.E.D.

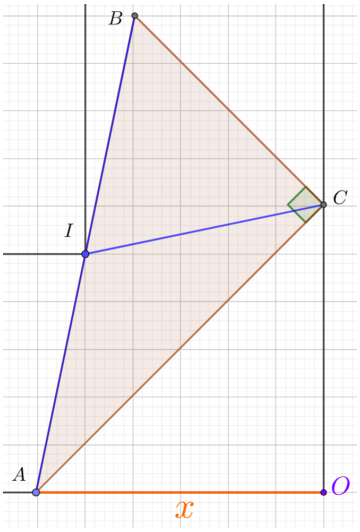


Figure 16: The right triangle after a $45°$ rotation

Hence, for the biggest acute triangle to move around $\mathcal{H}$ there exists an equivalent right angle triangle that can move around $\mathcal{H}$, and we do not have to consider acute triangles anymore. Lets maximize the area of the right angle triangles after a $45°$ turn. Let $x = |AO|$ as shown on the figure 16.

In order to maximize the area of $\mathcal{T}$ we want $I \in \overline{AB}$ and $C \in R_1$. This condition will definitely include the biggest right triangle that can navigate the entire turn, but not all the triangles satisfying this condition will be able to do it. As A and C are symmetric with respect to IO, $\overline{IA} = \overline{IC}$. As $\triangle ABC$ is right angle and $\overline{IA} = \overline{IC}$, I is the midpoint of $\overline{AB}$. Lets trace $\overline{IO}$ and $\overline{IM}$ such that $\overline{IM} \parallel \overline{AC}$. M_1 and M_2 are the intersections of $\overline{IO}$ with $\overline{AC}$ and $\overline{IM}$ with $\overline{BC}$ respectively. The figure 17 shows what we have done.

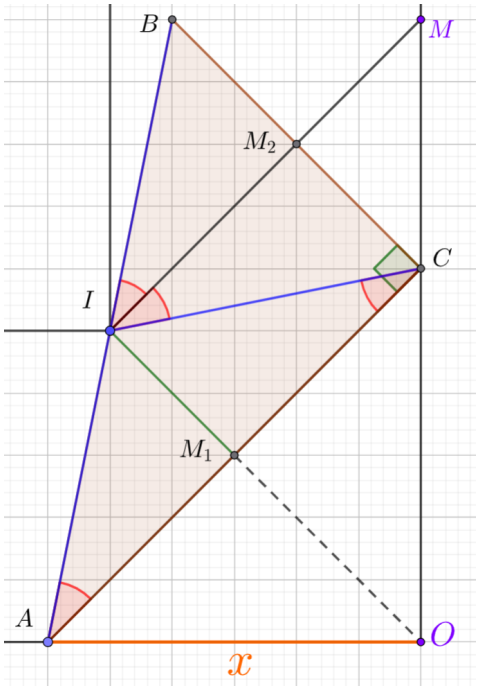


Figure 17: Properties of the right angle $\mathcal{T}$ sofas

Observe that

$$\begin{cases} \angle IAM_1 = \angle ICM_1 = \angle CIM_2 = \angle BIM_2, \\ \angle IM_1A = \angle IM_1C = \angle CM_2I = \angle BM_2I = 90°, \\ \overline{AI} = \overline{IB} = \overline{IC}. \end{cases}$$

Since the four triangles have the same length hypotenuse from point I and have equal corresponding angles,

$$\triangle IAM_1 = \triangle ICM_1 = \triangle CIM_2 = \triangle BIM_2$$

Consequently,

$$\text{Area}(\triangle ABC) = 4 \cdot \text{ Area}(\triangle IAM_1).$$

Since $\overline{AC} = \sqrt{2}x$, $\overline{AM_1} = \overline{OM_1} = \dfrac{\sqrt{2}}{2}x$. And as $\overline{IO} = \sqrt{2}$, the height of $\triangle IAM_1$ is

$$\overline{IM_1} = \overline{IO} - \overline{OM_1} = \sqrt{2} - \frac{\sqrt{2}}{2}x.$$

Therefore,

$$\begin{aligned}
\text{Area}(\triangle ABC) &= 4 \cdot \frac{1}{2} \cdot \frac{\sqrt{2}}{2}x \left(\sqrt{2} - \frac{\sqrt{2}}{2}x\right) \\
&= \sqrt{2}x \cdot \sqrt{2}\left(1 - \frac{x}{2}\right) \\
&= x(2 - x).
\end{aligned}$$

The maximum area possible of $\triangle ABC$ is the maximum value of the function $f(x) = x(2-x)$ when $x > 0$. As $x = 0$ and $x = 2$ are the roots of this concave down parabola, the maximum is found at $x = \dfrac{0+2}{2} = 1$. Substituting $x = 1$ back to the function, we get that the maximum area of the right angle triangle $\triangle ABC$ is $f(1) = 1 \cdot (2 - 1) = 1$. This is exactly the sofa we found for the lower bound.

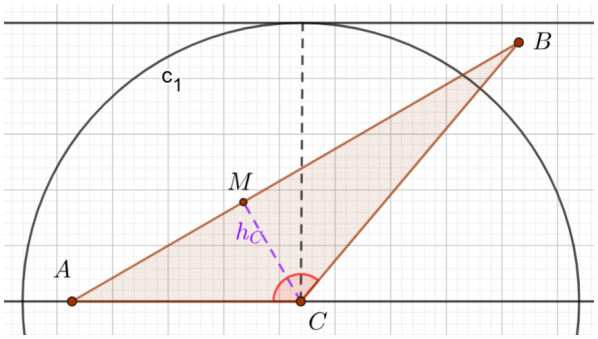


Figure 18: Height from C is smaller or equal to the radius of c_1

Finally, if $\mathcal{T}$ is obtuse on C, vertex C has one vetex of $\mathcal{T}$ on the left and one on the right. By claim 2, the height on C h_c must be smaller or equal to 1 as shown on the figure 18. Therefore, we can draw a translated and rotated triangle $\triangle A'B'C'$ of the triangle $\triangle ABC$, with $A'B'$ lying on R_4. The transformed triangle $\triangle A'B'C'$, shown on the figure 19 satisfies case 2. The obtuse triangle on A, is symmetric to the obtuse triangle on C. By the Remark 1, there is no need in considering both cases. Thus, the biggest area sofa $\mathcal{T}$ satisfying case 1, which cannot be transformed into case 2, is the right triangle with area 1.

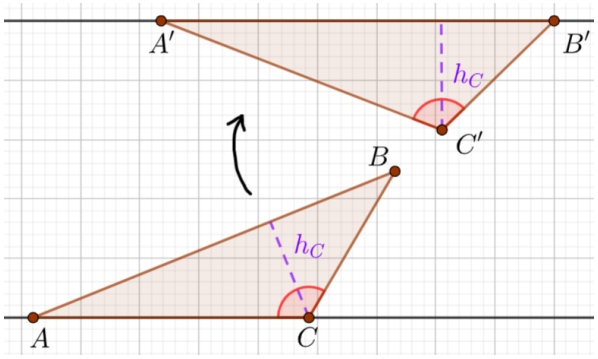


Figure 19: Transformation of the obtuse triangle

3.3 Case 2

We have narrowed all the candidates for the biggest sofa $\mathcal{T}$ to the ones satisfying case 2. For this case, lets once again rotate $\mathcal{T}$ $45°$, so it lies totally inside $T_1 = \triangle ONM$ with $\overline{BC}$ lying on $\overline{ON}$. If $\mathcal{T}$ has an obtuse angle it must be the angle $\angle BAC$ as if not BC would not be the longest side. This is shown on the figure 20.

Claim 4. *If triangle $\mathcal{T} = \triangle ABC$ is not isosceles, and it makes the turn, then the triangle $\triangle A'BC$ will also make the turn, where $\overline{A'B} = \overline{A'C}$ and $h_A = h'_A$.*

Proof

Consider only $\mathcal{T}$ that have $\angle BCA \geq \angle CBA$. If it does not, consider the reflection of $\mathcal{T}$ across h_A without loss of generality. Now consider the quadrilateral $Q = BCAA'$. If $\mathcal{T}$ can finish the $45°$ counter clockwise rotation, $\overline{BC}$ will end the rotation lying on R_3. In that

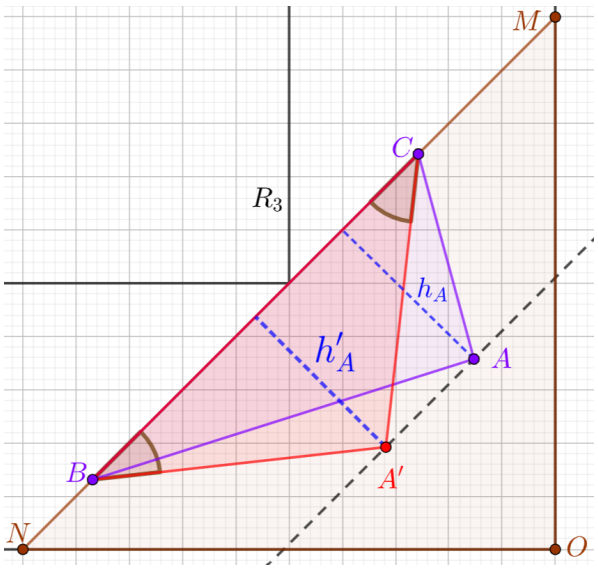


Figure 20: $\triangle A'BC$

case, Q will also be able to finish the turn, because during this turn $x(A') \leq x(A)$ so A' stays inside of $\mathcal{H}$. If Q can finish the last $45°$ rotation, the triangle $\triangle A'BC$ can also finish it. But $\triangle A'BC$ can also navigate the first $45°$ of rotation, as it is symmetric to the second half.

Q.E.D.

3.3.1 Analytical approach

Now, we only need to analyze the isosceles sofas $\mathcal{T}$ that begin the turn with $\overline{BC}$ lying on R_4 and finish it with $\overline{BC}$ lying on R_3. However, we can only consider the first $45°$ rotation, as the second half of the rotation will be symmetric to the first one with respect to OI. During this rotation, I always stays on BC and C always stays on R_1 as shown on the figure 21.

Lets define $O = (0, 0)$ as the point of origin and M as the midpoint of $\overline{BC}$. Also, r will be $\overline{MC}$ and h will be $\overline{OC}$. During the turn, h gets bigger, but r stays constant. Before starting the $45°$ rotation $BC \parallel R_4$ and $B \in R_4$, so $\overline{OC} = h = 1$. And at the end of the rotation, $\angle MCO = \angle MOC = 45°$, so $h = 2$. Hence, $h \in [1, 2]$. From the initial position of $\mathcal{T}$, we deduce that $h_A \leq 1$. We know that it is unnecessary to consider $\mathcal{T}$ with area smaller than 1, as it is the lower bound we have found. Therefore, we will consider $r \geq 1$. Similarly, from the equation 1, we know that any segment must be smaller or equal to $2\sqrt{2}$ in order to complete the turn. Consequently, $r \in \left[1, 2\sqrt{2}\right]$.

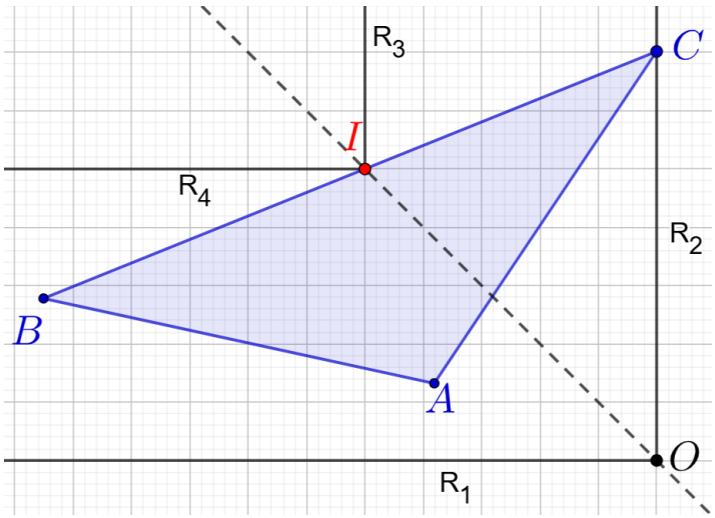


Figure 21: The movement of $\mathcal{T}$

Lets define the function $A(h, r)$, as h and r are independent. For the moment think of r as a fixed variable treating the function as $A(h)$. This function calculates for each h the area of the biggest isosceles triangle we can build that fits inside $\mathcal{H}$. By doing so, A will lie on R_2. as shown on the figure 22.

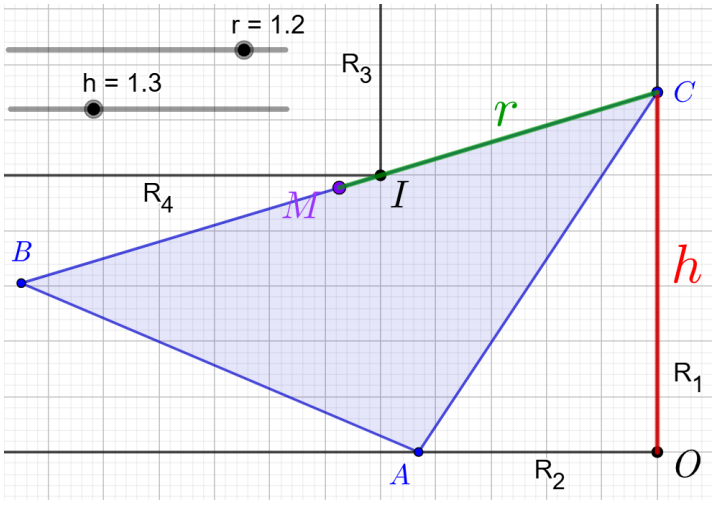


Figure 22: Isosceles triangle for $r = 1.2$ and $h = 1.3$

For a fixed r, we want to find the biggest triangle that fits inside $\mathcal{H}$ for all values of h. Lets look at the smallest triangle $\mathcal{T}_m$ that we draw, with area $\min(A(h))$. Notice, that $\mathcal{T}_m$ will fit inside $\mathcal{H}$ for all values of h, as it is the smallest of the triangles we draw. At the same time, $A(h)$ computes the area of the biggest isosceles triangle that fits inside $\mathcal{H}$. Hence, if we draw a slightly bigger isosceles triangle than $\mathcal{T}_m$, its area would be bigger than $\min(A(h))$ and it

would not not fit inside $\mathcal{H}$ for that angle. Thus, $\mathcal{T}_{max} = \mathcal{T}_m$ is the biggest triangle that can move around $\mathcal{H}$.

Property 1

Find the equation of the normal line N_1 to another line L_1 with slope a at the point O. The slope of $N_1 = \dfrac{-1}{a}$ and N_1 passes through (x_o, y_o) so the equation of N_1 is
$y = \dfrac{-1}{a}(x - x_o) + y_o$.

Strategy

1. Find the equation of $L_1 = MC$.
2. Compute the equation of $L_2 = MA$.
3. Find $\overline{MA}$.
4. Deduce the formula for the area of $\mathcal{T}$.
5. Prove that $A(h, r)$ is strictly increasing on r.
6. Find the area of $\mathcal{T}_m$ for the biggest value of r.

The equation of L_1

As we may observe, $C = (0, h)$ and $I = (-1, 1)$. So the equation of L_1 is

$$y - y_C = \frac{y_C - y_D}{x_C - x_D}(x - x_C) = y - h = \frac{h - 1}{0 - (-1)}(x - 0) = x(h - 1) \tag{2}$$

Let c_1 be a circle with center C and radius r. To satisfy $\overline{MC} = r$ we need the line L_1 to intersect c_1 at M as shown on the image 23.

The equation of the circle c_1 is $x^2 + (y - h)^2 = r^2$, and by the formula 2 the equation of L_1 is $y - h = x(h - 1)$. As they intersect at $M = (x_M, y_M)$, we obtain the following system

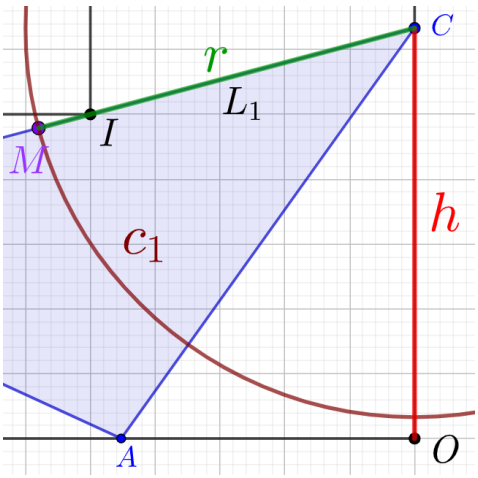


Figure 23: Intersection M between L_1 and c_1

of equations

$$\begin{cases} y_M - h = x_M(h-1) \\ x_M^2 + (y_M - h)^2 = r^2 \end{cases} \implies \begin{cases} (y_M - h)^2 = x_M^2(h-1)^2 \\ (y_M - h)^2 = r^2 - x_M^2 \end{cases}$$

Lets isolate x_M^2

$$x_M^2(h-1)^2 = r^2 - x_M^2 \implies x_M^2 = \frac{r^2}{(h-1)^2+1}$$

x_M is negative, so

$$x_M = -\frac{r}{\sqrt{(h-1)^2+1}}$$

Now lets substitute x_M^2 back to the second equation and isolate y_M

$$(y_M - h)^2 = r^2 - \frac{r^2}{(h-1)^2+1} = r^2\left(1 - \frac{1}{(h-1)^2+1}\right)$$

$y_M \leq h$ so

$$(y_M - h) = -r\frac{1}{\sqrt{(h-1)^2+1}} \implies y_M = h - r\frac{1}{\sqrt{(h-1)^2+1}}$$

Lets define $t = \sqrt{(h-1)^2+1}$ as it is a common factor, and simplify both expressions.

$$\begin{cases} x_M = \dfrac{-r}{t} \\ y_M = h - \dfrac{r(h-1)}{t} \end{cases}$$

The equation of L_2

For the next step we need to find the equation of L_2, the normal line to L_1 at M. By property 1, the equation of L_2 is $y = \dfrac{-1}{a}(x - x_M) + y_M$ where a is the slope of L_1 which from the formula 2 is $h-1$. As L_2 passes through A, we can plug in $A = (x_A, y_A)$. $y_A = 0$, so the only unknown is x_A. L_2 becomes

$$0 = \frac{-1}{h-1}(x_A - x_M) + y_M \implies (x_A - x_M) = y_M(h-1) \implies x_A = y_M(h-1) + x_M$$

Plugging in (x_M, y_M):

$$x_A = \left(h - \frac{r(h-1)}{t}\right)(h-1) - \frac{r}{t} = h(h-1) - \frac{r(h-1)^2 + r}{t}$$

$$= h(h-1) + \frac{rt^2}{t} = h(h-1) + rt$$

The distance $\overline{MA}$

We have found the coordinates of M and A. Now, lets find the distance between M and A shown on the figure 24. We know that

$$\overline{MA} = \sqrt{(x_A - x_M)^2 + (y_A - y_M)^2} \implies (\overline{MA})^2 = (x_A - x_M)^2 + (y_A - y_M)^2$$

Lets look at $y_A - y_M$

$$y_A - y_M = 0 - \left(h - \frac{r(h-1)}{t}\right) = -h + \frac{r(h-1)}{t}$$

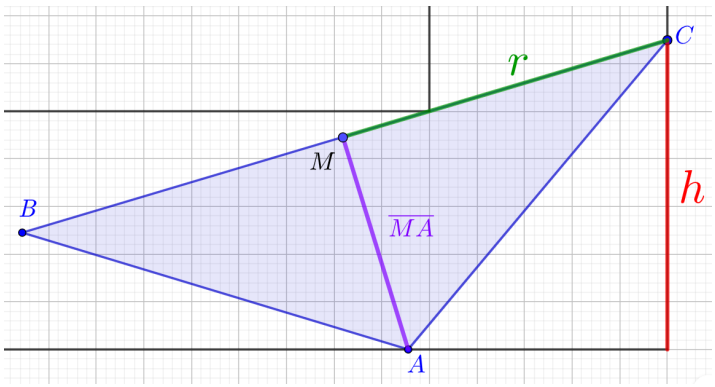


Figure 24: The segment $\overline{MA}$

Now, lets simplify $x_A - x_M$ to get a common factor with $y_A - y_M$

$$x_A - x_M = -rt + h(h-1) + \frac{r}{t}$$

The factor $\frac{r}{t} - rt$ can be simplified:

$$\frac{r}{t} - rt = r\left(\frac{1}{t} - t\right) = r\frac{1-t^2}{t} = r\frac{1-(1+(h-1)^2)}{t} = \frac{-r(h-1)^2}{t}$$

Hence

$$x_A - x_M = (h-1)h - \frac{r(h-1)^2}{t} = (h-1)\left(h - \frac{r(h-1)}{t}\right)$$

Lets simplify $(\overline{MA})^2$

$$\begin{aligned}(\overline{MA})^2 &= \left((h-1)\left(h - \frac{r(h-1)}{t}\right)\right)^2 + (-1)^2\left(h - \frac{r(h-1)}{t}\right)^2 \\ &= \left(h - \frac{r(h-1)}{t}\right)^2\left((h-1)^2 + 1\right)\end{aligned}$$

The second factor is t^2, so

$$\overline{MA} = \left(h - \frac{r(h-1)}{t}\right)t = ht - r(h-1) = h\sqrt{1+(h-1)^2} - r(h-1)$$

We got $\overline{MC}$ and $\overline{MA}$. We are ready to deduce $A(h)$,

$$A(h) = \overline{MC} \cdot \overline{MA} = r \cdot \left(h\sqrt{1 + (h-1)^2} - r(h-1) \right)$$

$$= rh\sqrt{1 + (h-1)^2} - r^2(h-1)$$

3.3.2 Optimization

Now, lets think of h as a constant and r as a variable viewing our function as $A(r)$. After building a Geogebra simulation, I have noticed that the area increases when I increase r.

Claim 5. *$A(r)$ is monotonously increasing on r on the domain $r \in [1, \sqrt{2}]$ and $h \in [1, 2]$.*

Proof

This claim is equivalent to proving that $\dfrac{\partial}{\partial r}(A(h, r)) \geq 0$.

$$\frac{\partial}{\partial r}(A(h, r)) = h\sqrt{1 + (h-1)^2} - 2r(h-1)$$

This expression is linear on r so when r grows, the rhythm at which r is growing decreases. In other words, $\dfrac{\partial}{\partial r}(A(h, r))$ decreases. Hence, to verify whether $A(h, r)$ monotonously increases or not, we need to verify whether the partial derivative is positive or negative for the biggest value of r. This value is $r = \sqrt{2}$, so we need to verify that the following expression is always positive on the domain:

$$\frac{\partial A}{\partial r}(h, \sqrt{2}) = h\sqrt{1 + (h-1)^2} - 2\sqrt{2}(h-1)$$

We want to see that

$$h^2(1 + (h-1)^2) \geq 8(h-1)^2 \implies h^2(1 + (h-1)^2) - 8(h-1)^2 \geq 0$$

The expression is equivalent to the following polynomial:

$$P(x) = h^4 - 2h^3 - 6h^2 + 16h - 8$$

Evaluating $P(x)$ at $x = 2$ using Ruffini's rule, we see that $x = 2$ is a double root of this polynomial.

	1	−2	−6	16	−8
2		2	0	−12	8
	1	0	−6	4	0

	1	0	−6	4
2		2	4	−4
	1	2	−2	0

So $P(x)$ can be simplified to

$$P(x) = (h-2)^2(h^2 + 2h - 2)$$

The factor $(h-2)^2$ is always positive, so we need to see that

$$(h^2 + 2h - 2) \geq 0$$

for $h \in [1, 2]$. Lets obtain the roots of $h^2 + 2h - 2 = 0$ using the quadratic formula:

$$x = \frac{-2 \pm \sqrt{2^2 - 4 \cdot 1 \cdot (-2)}}{2} = -1 \pm \frac{\sqrt{4(1+2)}}{2} = -1 \pm \sqrt{3}$$

$$\sqrt{3} < \sqrt{4} = 2 \implies -1 + \sqrt{3} < 1$$

Since we are only interested in the domain $h \in [1, 2]$ and the parabola $h^2 + 2h - 2$ is concave up, we have

$$h > -1 + \sqrt{3} > -1 - \sqrt{3}$$

Therefore, on our domain,

$$h^2 + 2h - 2 > 0 \implies P(x) \geq 0 \implies \frac{\partial}{\partial r}(Area(h, r)) \geq 0$$

The area of the biggest isosceles $\mathcal{T}$ is monotonously growing on r.

Q.E.D.

3.4 Solution

We know that $A(h, r)$ is monotonously increasing on r. In order to find the area of the biggest area sofa $\mathcal{T}_{max}$ that makes the turn, we need to minimize the function

$$A(h, \sqrt{2}) = \left(\sqrt{2}h\sqrt{1 + (h-1)^2} - (\sqrt{2})^2(h-1)\right)$$

As this expression only has fractions and roots of polynomials, Wolfram Alpha found the minimum on the domain $h \in [1, 2]$:

$$\min(A(h, \sqrt{2})) = \text{Area}(\mathcal{T}_{\max}) = \boxed{\frac{3}{2}\left(\frac{\sqrt[3]{4+3\sqrt{2}}}{2^{2/3}} - \frac{1}{2\sqrt[3]{4+3\sqrt{2}}}\right)} \approx 1.3194$$

$$\text{for} \quad h = 1 - \frac{1}{2\sqrt[3]{1+\sqrt{2}}} + \frac{1}{2}\sqrt[3]{1+\sqrt{2}} \approx 1.2980$$

Notice, that the biggest triangle for case 2 is bigger than the biggest triangle for case 1. Therefore, this is the biggest triangle $\mathcal{T}_{max}$ that can move around $\mathcal{H}$. The movement of this sofa $\mathcal{T}_{max}$ that we have followed, is shown in 4 snaps of the figure 25. Only the first half of the rotation is shown, as the other one is symmetric by Remark 1.

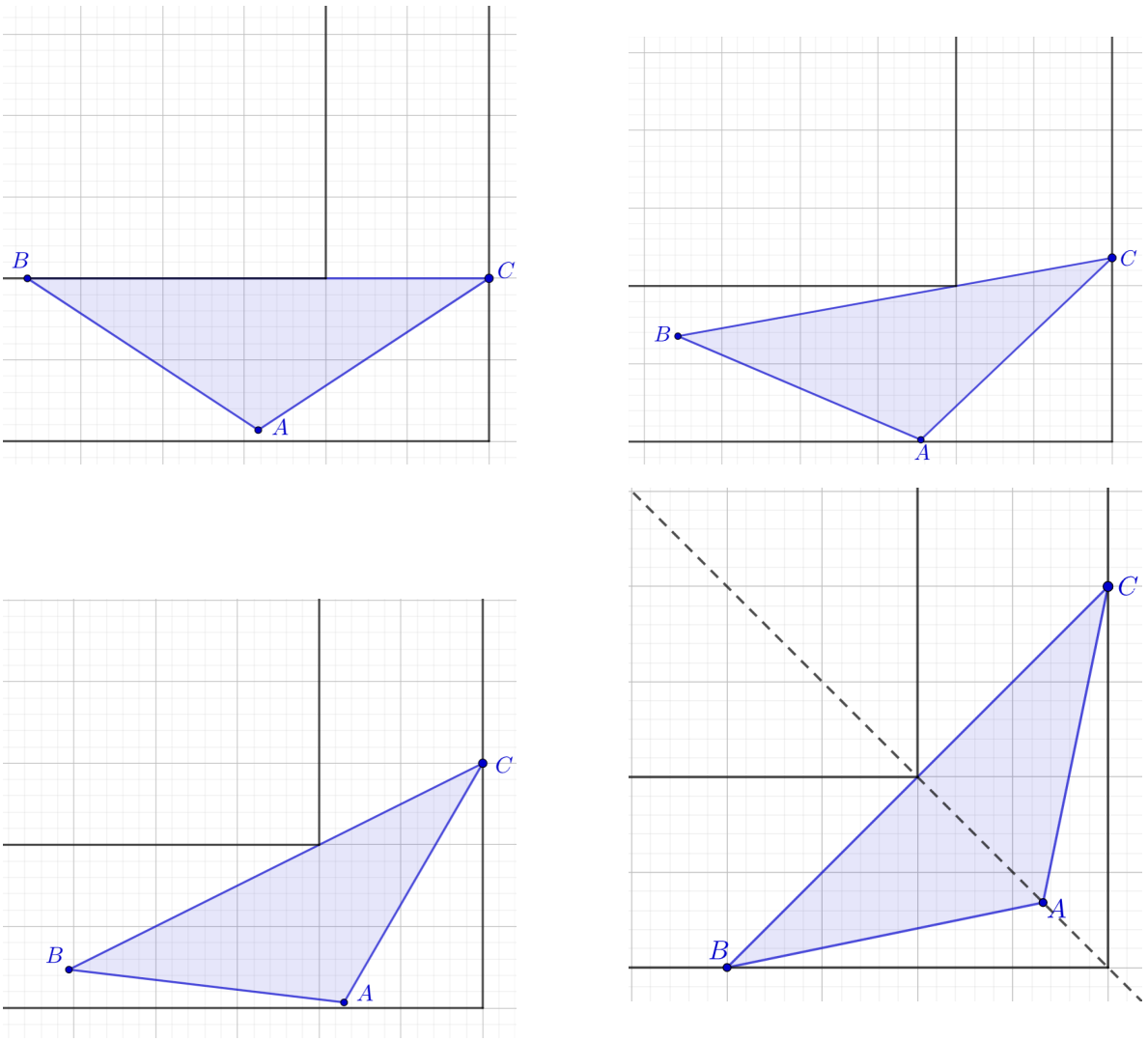


Figure 25: The movement of the sofa $\mathcal{T}_{max}$ inside $\mathcal{H}$

4 Simulation for the biggest triangle

For now, we have been rotating sofas $\mathcal{S}$ around $\mathcal{H}$, while $\mathcal{H}$ did not move. This strategy is useful when the equations that describe the boundaries of $\mathcal{S}$ are simple enough to express algebraically. This was the case for sofas $\mathcal{D}$ and $\mathcal{T}$. But in case a sofa $\mathcal{S}$ has an irregular shape, we need to change the perspective. We will move the corridor $\mathcal{H}$ around a fixed sofa $\mathcal{S}$. The area enclosed inside $\mathcal{H}$ throughout all the rotation of $\mathcal{H}$ is a valid sofa $\mathcal{S}$ to complete the turn. This approach is handy when you want to find a valid sofa, without defying its exact shape. Gerver [3] expanded this method to come up with a candidate for the biggest shape $\mathcal{S}$ to move around $\mathcal{H}$. Instead of moving $\mathcal{S}$, he thought of the problem as continuously

translating and rotating $\mathcal{H}$ through $90°$ by the Assumption 1. The area which was inside $\mathcal{H}$ throughout all the rotation of $\mathcal{H}$ can complete the turn. An example is shown in figure 26.

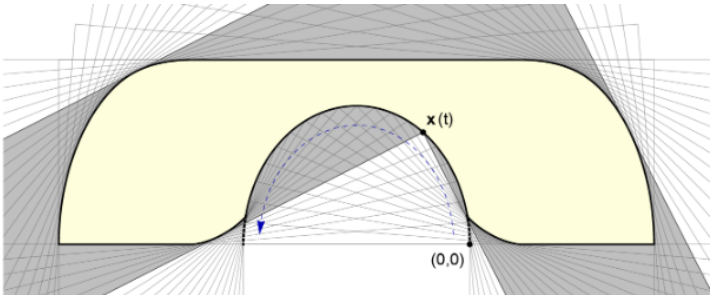


Figure 26: An example of the area enclosed inside all the translations of $\mathcal{H}$. The figure is taken from Romik's paper [7]

The strategy to find Gerver's sofa, is explained in the Appendix B. Now, lets use Gerver's idea of taking N snaps of the rotating corridor to simulate the biggest triangle $\mathcal{T}_{max}$.

Relevant restriction

We shall remember that this algorithm only simulates the rotation of the sofa $\mathcal{S}$, and not the straight movement in one direction without rotation. For example, our program could create the sofa from the figure 27. This sofa can rotate $90°$ inside $\mathcal{H}$ but cannot move vertically or horizontally.

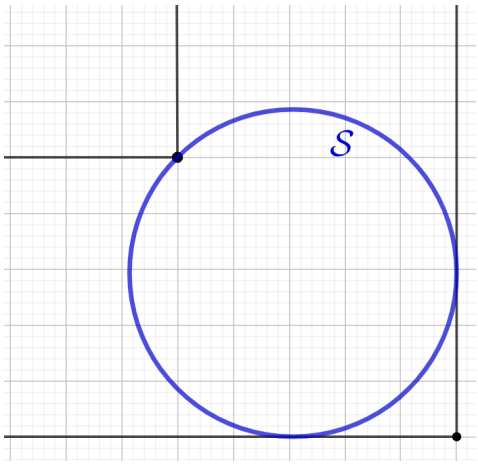


Figure 27: Counterexample for the width condition

Therefore, after each simulation we need to verify whether the resulting shape fits inside a strip of width 1. The figure 28 shows a valid sofa S.

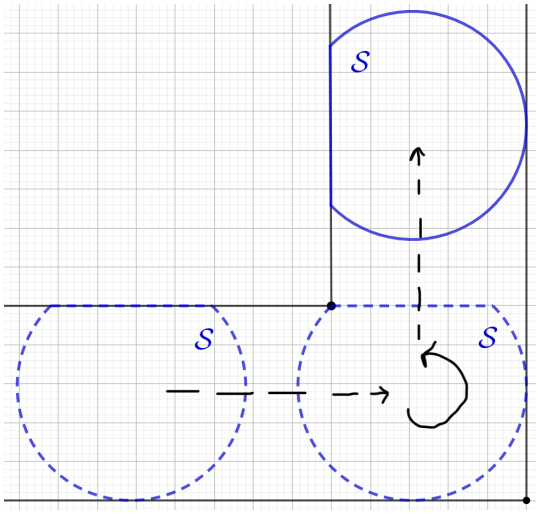


Figure 28: Sofa from the figure 27, after applying the restriction.

Now, lets build $\mathcal{T}_{max}$ using this discrete method. As the sofa will stay fixed, the corridor $\mathcal{H}$ will rotate and translate. Lets define $\mathcal{H}_0$ as the corridor at its initial position and $\mathcal{H}_i$ as the copy of $\mathcal{H}_0$ after i rotations. This way, $\mathcal{H}_N$ is the last snap of the corridor. $\overline{BC} = 2\sqrt{2}$ and the inner corner I stays in contact with $\overline{BC}$ throughout the turn. For this reason, it will move $2\sqrt{2} - 2$ units to the left as shown on the figure 29. We can also observe on this image $\mathcal{H}_0$ and $\mathcal{H}_N$.

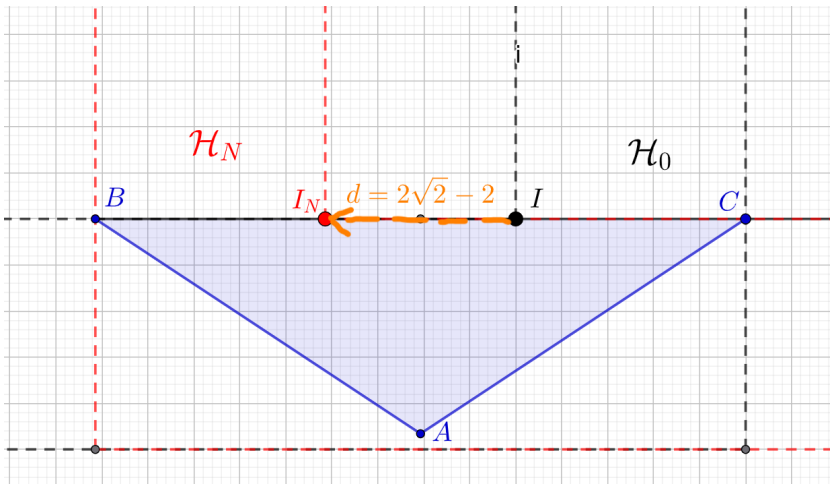


Figure 29: Inner corner path

Now, lets look at a snap of the corridor $\mathcal{H}_i$ at a random moment during the turn. It is shown on the figure 30. The corner I_i has moved dx units from its initial position I on $\mathcal{H}_0$. Lets define α as how much has rotated $\mathcal{H}_i$ with respect to its initial position $\mathcal{H}_0$. The width

of the corridor is 1, so $\overline{I_iF} = \overline{IC} = 1$. For this reason,

$$\alpha = \arcsin\left(\frac{1}{1+dx}\right)$$

Thus, $\mathcal{H}_i$ differs from $\mathcal{H}_0$ by a rotation of α units clockwise and a translation of dx units to the left. Both variables depend only on dx, so we now the exact position of each corner $\mathcal{H}_i$.

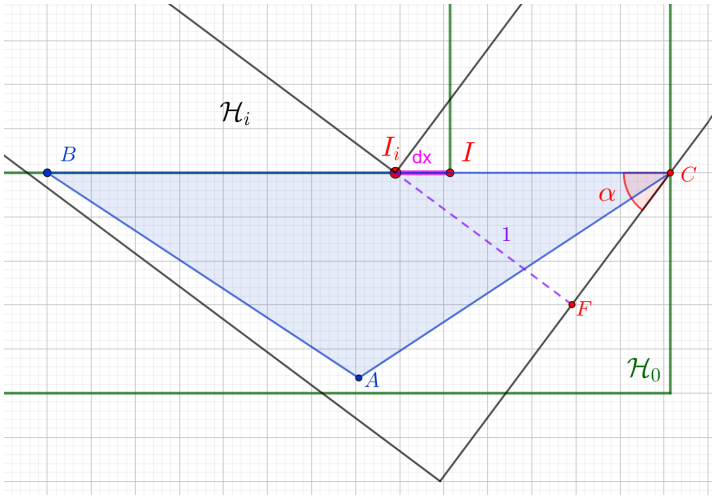


Figure 30: Finding α

In the appendix C I have written a code which draws N copies $\mathcal{H}_i$ of the initial corridor $\mathcal{H}_0$ and finds the intersection between all these copies. By running the simulation for $N = 30$ as shown on the figure 31, we get the following shape with an approximate area of 1.5515 units.

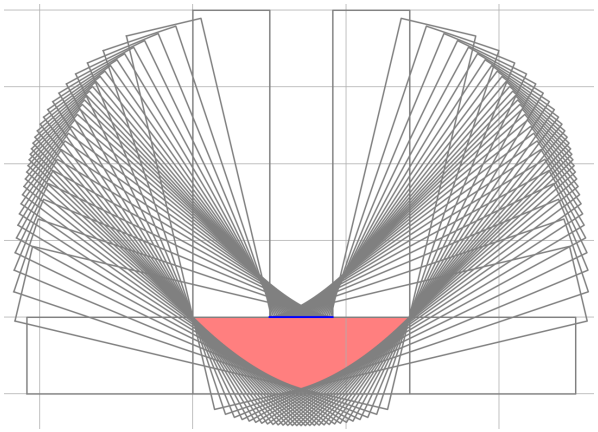


Figure 31: Simulation result of extended $\mathcal{T}_{max}$

This shape fits in a horizontal strip of width 1, and therefore it can complete the turn around the corner. If we look at the 3 vertices of this shape, we may notice that connecting them we obtain $\mathcal{T}_{max}$. This is shown on the figure 32. Therefore, $\mathcal{T}_{max}$ is found inside this shape, so $\mathcal{T}_{max}$ can also navigate around the corner $\mathcal{H}$.

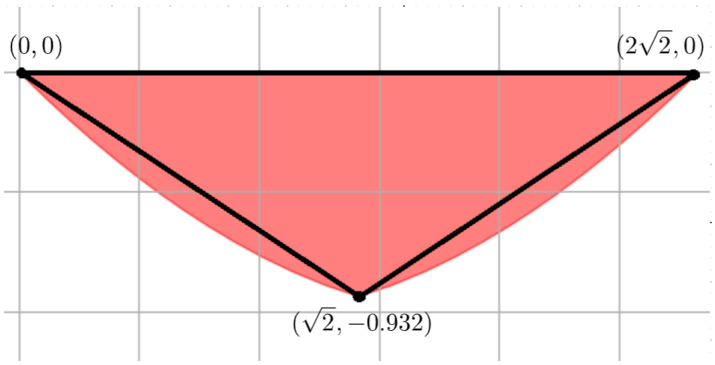


Figure 32: Finding $\mathcal{T}_{max}$ inside of the intersection

The simulation has revealed a different approach to verify that $\mathcal{T}_{max}$ can navigate around the corner. On one hand, this approach is less precise than the analytical geometry approach as it is an approximation. But on the other hand, this approach has revealed a slightly bigger shape than $\mathcal{T}_{max}$ that can also navigate around the corner, extending the triangle case. It is a powerful instrument to deduce optimal shapes, but not to prove them.

5 Conclusion

While we were searching for the biggest sofa $\mathcal{T}_{max}$, we had to deal with nontrivial problems which we carefully simplified into smaller problems we could think about and attempt solving. The key idea was to rotate the initial triangle $45°$, as everything became symmetric about IO. From that point on, we controlled the orientation of $\mathcal{T}$ and deduced that $\mathcal{T}_{max}$ is an isosceles triangle. The longest side $\overline{BC}$ of $\mathcal{T}_{max}$ is equal to the longest segment $\mathcal{D}$ that can be moved around the corner. The y coordinate of the third vertex A is irrational and needed a lot of analytical geometry and calculus to be found. This reflects on the difficulty to describe the continuous movement of a shape that is rotating and translating simultaneously. Using a completely different approach, we simulated the rotation of the corner $\mathcal{H}$ around $\mathcal{T}_{max}$ to verify our solution. In order to do that, we had to find the exact translations and rotations $\mathcal{H}$ followed. The agreement between both approaches we have used suggests that the sofa $\mathcal{T}_{max}$ is the biggest triangle that can move around a $90°$ corner.

If the simulation could be expressed mathematically, it could be more efficient than the analytical approach. However, our simulation was an approximation because we took snaps of the rotation of $\mathcal{H}$, not considering it continuous. In our paper, we prove that the optimal sofa must rotate at least $45°$. Nevertheless, we have assumed that the optimal sofa $\mathcal{T}_{max}$ could rotate exactly $90°$ to complete the turn.

6 Bibliography

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A Hammersley's minimum angle of rotation

Claim 6. *The sofa with biggest area must rotate at least* $45°$ *to make the turn.*

Proof

Let S_1 be a sofa that can navigate through the turn rotating only through an angle of α. Let A_s denote the area of S_1. Before rotating, S_1 was located inside a horizontal strip of width 1. After making the turn, the horizontal strip has rotated $90°$ and S_1 has rotated α. Thus, S_1 must be contained in the intersection of a strip of width 1 and the strip rotated $90° - \alpha$. This intersection is the purple parallelogram shown on the figure 33.

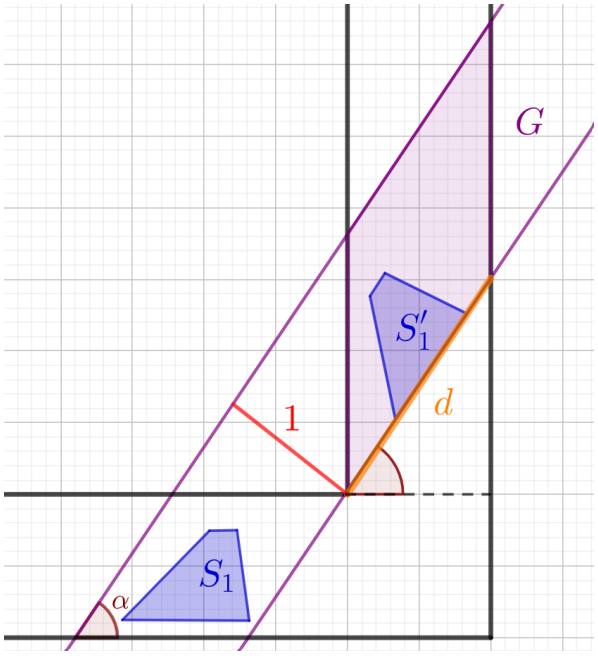


Figure 33: α rotation of $\mathcal{S}$

The area of this parallelogram is

$$A_\alpha = d \cdot h = \frac{1}{cos(\alpha)} \cdot 1$$

And we know for any S_1 that

$$A_s \leq A_\alpha \implies A_s \leq \sec(\alpha)$$

For example, we know that the semicircle with radius 1 can make the turn. Thus, the semicircle is a valid shape for S_1. Its area is $\frac{\pi}{2}$. Therefore,

$$\frac{\pi}{2} \leq A_\alpha = \frac{1}{\cos(\alpha)} \implies \frac{2}{\pi} \geq \cos(\alpha)$$

As the domain of α is $[0°, 90°]$, α is inside the domain of the function $(f(x) = \arccos(x))$. $f(x)$ is a strictly decreasing function, for that reason,

$$\arccos\left(\frac{2}{\pi}\right) \approx 50.46° \leq \arccos(\cos(\alpha)) = \alpha$$

Therefore, it is necessary to rotate more than $45°$ to move a shape with area bigger or equal to the area of the semicircle around the corner.

Q.E.D.

B Gerver's sofa

In 1992, Gerver[3] published a shape he claimed to be the biggest one, although without a demonstration. From all the possible rotations and translations of $\mathcal{H}$, Gerver found a methodology to draw a candidate which enclosed the biggest $\mathcal{S}$. Lets assume we only take 3 snapshots during the rotation of $\mathcal{H}$ and try to find the biggest intersection between these snaps. On the figure 34, we can the 3 snaps: the corner rotated $0°$ called $\mathcal{H}_0$, the corner rotated $45°$ called $\mathcal{H}_1$ and the corner rotated $90°$ called $\mathcal{H}_2$. We can observe that the area enclosed by the 3 corners is not maximal. We can move $\mathcal{H}_1$ corridor south-east and paint green the area of the trapezoid that we gain and red the area of the trapezoid that we loose. Notice, that the green segment is longer than the red segment. For that reason, the green area that we gain is bigger than the red area we loose.

This optimization can be carried out until the red and the green segment have the same length, as afterwards we would loose area. We can apply this idea whenever the opposite sides are not equal, and gain area. Therefore, we deduce that the maximum area polygon

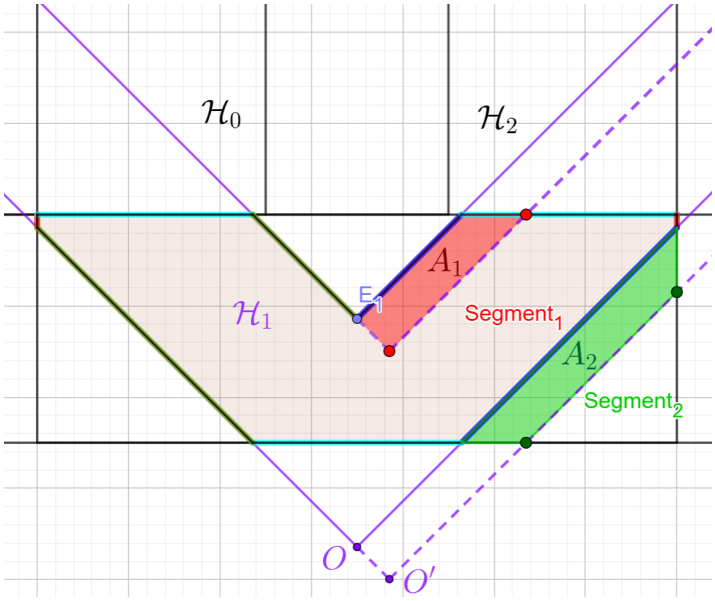


Figure 34: Bigger area polygon. Image inspired from this source [8]

will have all the opposite sides of equal length, as shown on the figure 35. The polygons that share this property are called balanced polygons.

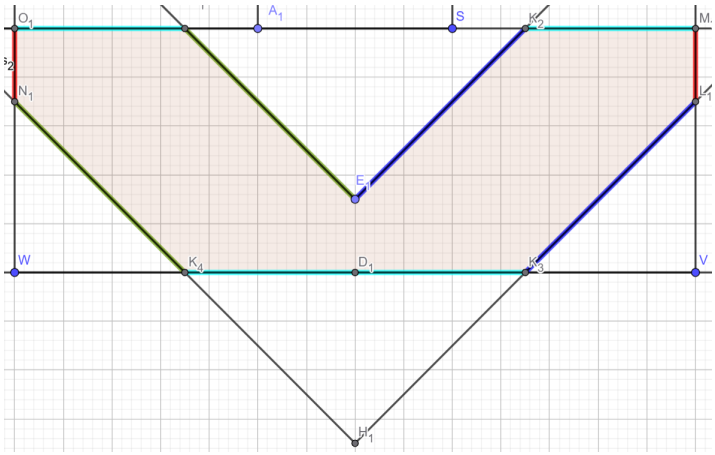


Figure 35: Balanced polygon for the 3 snaps of $\mathcal{H}$

Lets call N the number of snaps we take to $\mathcal{H}$ during the turn. As we increase N, the movement between the rotated corridors becomes continuous. For this reason, when $N \to \infty$ the resultant sofa $\mathcal{S}$ will be able to make the turn. In order to make the biggest sofa $\mathcal{S}$ that can move around $\mathcal{H}$, we should draw the balanced polygon for the intersection of N hallways as $N \to \infty$.

C Python code for the simulation

```python
import math
import matplotlib.pyplot as plt
from shapely.affinity import rotate, translate
from shapely.geometry import Polygon
from functools import reduce

def InitialCorridor():
    # Define the length and width of the corridor
    Corridor = Polygon([(0, 5), (0, 0), (5, 0), (5, 1), (1, 1), (1, 5)])
    return Corridor

def RotateAndTranslate(Corridor, Displacement):
    RotatedCorridors = [Corridor]  # start with the original corridor
    #Define the pivot point
    pivot_x = 1
    pivot_y = 1
    dx = Displacement / N #Define the mini step

    for i in range(N):
        #For each iteration, update the pivot, translate and rotate the
            corridor.
        pivot_x += dx
        if(i<N//2): #First 45 degree rotation
            d_alpha = math.acos(1/pivot_x) # total angle H_0 must rotate
        elif(i==N-1):
            d_alpha = math.pi/2 #We cannot divide by 0, therefore the
                last iteration is an elif.
        else:
            d_alpha = math.pi/2-math.acos(1/(Displacement+2-pivot_x)) #
                the symmetric case with respect to the 45 degree case.
        d_alpha = d_alpha*180/math.pi     #Change to degrees

```

```
30          #print(d_alpha) If you want to see each angle of rotation
31          #Translate and rotate
32          Corridor = translate(Corridor, xoff=pivot_x-1)
33          Corridor = rotate(Corridor, d_alpha, origin=(pivot_x, pivot_y))
34          #Add to the list of corridors
35          RotatedCorridors.append(Corridor)
36          Corridor=InitialCorridor() # As we only know how to calculate the
                total d_alpha,
37          #we must return to the initial corridor after each iteration
38
39      return RotatedCorridors
40
41  def intersectTwoPolygons(a, b): #Function we will use to calculate the
        intersection of all the rotated corridors.
42      return a.intersection(b)
43
44  def PlotCorridorsAndIntersection(RotatedCorridors, Displacement):
45      plt.figure()
46      plt.title("Turn the hallway!")
47      plt.xlabel("X")
48      plt.ylabel("Y")
49      plt.axis('equal')    # x:y=1:1
50      plt.grid(True)  #Add a grid
51      for Corridor in RotatedCorridors:
52          x, y = Corridor.exterior.xy  #Get the vertives of the polygon
                with the first one repeated.
53                                       #In order to plot the polygon
54          plt.plot(x, y, color='gray')
55
56      # find and print the area of the intersection
57      Intersection = reduce(intersectTwoPolygons, RotatedCorridors)
58      InterArea = Intersection.area
59
60      print("Area of intersection: ", Intersection.area)
```

```
    #Reduce recursively applies the function for all values in the list

    #plot the intersection
    x, y = Intersection.exterior.xy
    plt.fill(x, y, color='red', alpha=0.5, label='Intersection')



    #Plot the pivot path
    x=[1,1+Displacement]
    y=[1,1]
    plt.plot(x, y, color='blue', linewidth=2, label="Pivot path")
    plt.legend()
    plt.show()

#Number of turns we will apply to the initial corridor
N=30
#The length the inner corner I will displace in a straight line
d=2*math.sqrt(2)-2
Corridor=InitialCorridor()
RotatedCorridors=RotateAndTranslate(Corridor, d)
PlotCorridorsAndIntersection(RotatedCorridors, d)
```